



Figure 1: Add right-click menu options for media files using Caja Actions and FFmpeg bash scripts

with incorrect key frames. These errors can make conversions or editing difficult or impossible. FFmpeg can fix these errors if you instruct it to simply *copy* the audio and video streams to a new file.

```
ffmpeg -i duffy.ogv \
-c:v copy \
-c:a copy \
duffy-fixed.ogv
```

- **Rotate video:** Sometimes, the videos that people take from a mobile phone have been rotated 90 or 180 degrees. You can fix this by specifying a transpose filter, as follows. The transpose value is in multiples of 90.

```
ffmpeg -i tank.mp4 \
-vf "transpose=1" \
tank-rotated.mp4
```

- **Cut a part of a video:** To extract a part of a video, you need to specify the start time using the `-ss` option, and the duration using the `-t` option (use the hh:mm:ss format). If you don't want the file to be re-encoded, then 'copy' the streams, as shown below:

```
# start from 1 min 12 sec and go for 2 min 50 sec
ffmpeg -i tank.mp4 \
-ss 1:12 \
-t 2:50 \
tank-cut-recoded.mp4
```

```
ffmpeg -i tank.mp4 \
-c:v copy \
-c:a copy \
-ss 1:12 \
-t 2:50 \
tank-cut.mp4
```

- **Append a video to another video:** FFmpeg offers several methods for merging. The simplest is to use input videos that use the same codec and, preferably, the same resolution and frame rate. The following commands first create a text file containing the path names of the input files. The FFmpeg commands merge the files specified in the text file.

```
echo "file 'tank-cut-recoded.mp4'" > file-list.txt
echo "file 'tank.mp4'" >> file-list.txt
ffmpeg -f concat -i file-list.txt -c copy tank-mix.mp4
```

- **Perform two-pass encoding:** With two-pass encoding, you can get a higher quality video file with fewer artifacts. The data obtained from the first pass is recorded in a log file and then used in the second pass for a more efficient conversion.

```
ffmpeg -y -i duffy.ogv -f mp4 \
-pass 1 -passlogfile duffy-log-file \
-vcodec mpeg4 -s 320x240 -r 25 -b:v 512K \
-an /dev/null
```

```
ffmpeg -i duffy.ogv -f mp4 \
-pass 2 -passlogfile duffy-log-file \
-vcodec mpeg4 -s 320x240 -r 25 -b:v 512K \
-acodec libmp3lame -ar 44100 -b:a 96K \
duffy.mp4
```

Klaatu (of the GNU World Order podcast) once provided a great video-editing tip. First, down-sample (lower the bit rate with the same resolution) the input video and create your video edits. When the video editing project file is finalised, replace the down-sampled video file with the original, bigger video file. This way, the video editing process and your test renderings are faster. The final rendering can take hours to finish, but you do not waste much time waiting before that. You can use FFmpeg to down-sample the video.

FFmpeg automation

If you have Caja Actions or Nautilus Actions Configuration installed, then you can simply right-click media files to convert them. You will have to first write some BASH scripts that will automate the FFmpeg conversions. Nautilus Actions Configuration (GNOME 2/3) is available in most software repositories. My March 2016 article has steps for compiling Caja Actions (Mate). 

References

- [1] <http://ffmpeg.org/> "FFmpeg.org"
- [2] <http://trac.ffmpeg.org/wiki/CompilationGuide> "FFmpeg compilation guide"

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